第二章：行波法

一、一维波动方程的初值问题：d'Alembert公式

考虑无界弦（$x\in\mathbb{R}$）的自由振动（不加f）：

弦振动的物理定律：各质点的加速度

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u & (x,t)\in\mathbb{R}\times(0,\infty) \\ u(x,0)=\phi(x),\quad \partial_t u(x,0)=\psi(x), & x\in\mathbb{R}. \end{cases}$$

两个变元　二维

二阶方程：两个初值条件　初始状态

初始位置　初始速度

达朗贝尔公式：

$$u(x,t)=\frac{1}{2}[\phi(x+at)+\phi(x-at)]+\frac{1}{2a}\int_{x-at}^{x+at}\psi(s)ds.$$

⇒ 利用特征线法证明达朗贝尔公式：

把PDE转化为ODE $(u\to\bar{u})$

$$\partial_t^2 u - a^2\partial_x^2 u = 0 \Rightarrow \underbrace{(\partial_t + a\partial_x)}_{2a\partial_\xi}\underbrace{(\partial_t - a\partial_x)}_{-2a\partial_\eta}u=0.$$

通过坐标变换 → 类似于化标准型.

① constraint coordinate $(x,t)\mapsto(\xi,\eta)$

s.t.
$$\begin{cases}\partial_t + a\partial_x = 2a\partial_\xi \\ \partial_t - a\partial_x = -2a\partial_\eta\end{cases} \Rightarrow \begin{cases}\partial_t = a(\partial_\xi-\partial_\eta) \\ \partial_x = \partial_\xi+\partial_\eta\end{cases}$$

$$\begin{cases}\xi_x=1\\ \xi_t=a\end{cases},\ \boxed{\xi=x+at}\qquad \begin{cases}\eta_x=1\\ \eta_t=-a\end{cases},\ \boxed{\eta=x-at}$$

将二阶偏导项变为只有一个，从而得到一个ODE → 将$\bar{u}$（两个自变量）表示为F, G（一个自变量）etc. 的组合.

② If so $\bar{u}(\xi,\eta)=u(x(\xi,\eta),t(\xi,\eta))$ satisfies

$$\boxed{\partial^2_{\xi\eta}\bar{u}=0}$$

代入定解条件求解F, G…

[另：解特征方程可得到同样的结果]　⇒ 两族特征线

Take $v(\xi,\eta)=\bar{u}_\xi$, $v_\eta=0$, $v=v(\xi)=\bar{u}_\xi$

$\bar{u}_\xi$是只关于ξ的函数.

牛顿-莱布尼茨公式：$\bar{u}(\xi,\eta)-\underbrace{\bar{u}(0,\eta)}_{\text{只与}\eta\text{有关}}=\int_0^\xi \bar{u}_\xi(t,\eta)dt = h(\xi).$

$$\Longrightarrow \bar{u}(\xi,\eta)=h(\xi)+g(\eta)$$

$$\boxed{u(x,t)=h(x+at)+g(x-at)}\quad\text{(行波解).}$$

h(x)以a的速度向左平移：左行波　　g(x)以a的速度向右平移：右行波.

初值条件 $u(x,0)=\phi(x)$, $\partial_t u(x,0)=\psi(x)$.

$$\begin{cases} h(x)+g(x)=\phi(x) \\ a h'(x)-a g'(x)=\psi(x) \end{cases} \Rightarrow h(x)-g(x)=\frac{1}{a}\int_0^x \psi(s)ds + C.$$

$$\Rightarrow \begin{cases} h(x)=\frac{1}{2}\phi(x)+\frac{1}{2a}\int_0^x \psi + \frac{1}{2}C. \\ g(x)=\frac{1}{2}\phi(x)-\frac{1}{2a}\int_0^x \psi - \frac{1}{2}C. \end{cases}$$

$$u(x,t)=h(x+at)+g(x-at)$$

$$=\frac{1}{2}[\phi(x+at)+\phi(x-at)]+\frac{1}{2a}\int_{x-at}^{x+at}\psi(s)ds.$$

eg1. $$\begin{cases} u_{xx}-2u_{xy}-3u_{yy}-u_x-u_y=0 \quad (x\geq 0, y\geq 0). \\ u(0,y)=f(y) \quad (y\geq 0), \quad u_y(x,0)=e^{\frac{x}{4}} \quad (y\geq 0). \\ u(x,0)=0 \quad (x\geq 0). \end{cases}$$

特征方程 $\phi_x^2-2\phi_x\phi_y-3\phi_y^2=0$. $(\phi_x+\phi_y)(\phi_x-3\phi_y)=0$.

$\frac{dy}{dx}=1$ $\qquad$ $\frac{dy}{dx}=-3$

$\Rightarrow$ 特征线 $\xi=x-y$, $\eta=3x+y$.

$$\bar{u}(\xi,\eta)=u\left(\frac{\xi+\eta}{4}, \frac{-3\xi+\eta}{4}\right).$$

$$\bar{u}_\xi=u_x\cdot\left(\frac{1}{4}\right)+u_y\cdot\left(-\frac{3}{4}\right)$$

$$\bar{u}_\eta=u_x\cdot\left(\frac{1}{4}\right)+u_y\cdot\left(\frac{1}{4}\right)$$

$$\bar{u}_{\xi\eta}=\frac{1}{4}\cdot\left[u_{xx}\cdot\frac{1}{4}+u_{xy}\cdot\frac{1}{4}\right]-\frac{3}{4}\cdot\left[u_{xy}\cdot\frac{1}{4}+u_{yy}\cdot\frac{1}{4}\right]$$

$$=\frac{u_{xx}-2u_{xy}-3u_{yy}}{16}=\frac{u_x+u_y}{16}=\frac{\bar{u}_\eta}{4}.$$

即 $4\bar{u}_{\xi\eta}=\bar{u}_\eta$.

设 $\bar{u}_\eta=v(\xi,\eta)$, 则 $v_\xi=\frac{1}{4}v$, $\ln v=\frac{\xi}{4}+C$. 令 $\xi=0$, $C=\ln v(0,\eta)$

$$\therefore \underbrace{\bar{u}_\eta}_{\text{对}\eta\text{积分}}=v=e^C\cdot e^{\frac{\xi}{4}}=v(0,\eta)e^{\frac{\xi}{4}}=\bar{u}_\eta(0,\eta)e^{\frac{\xi}{4}}.$$

再由牛顿-莱布尼茨公式: $\bar{u}(\xi,\eta)-\bar{u}(\xi,0)=\int_0^\eta \bar{u}_\eta(\xi,t)dt=e^{\frac{\xi}{4}}\int_0^\eta \bar{u}_\eta(0,t)dt$

$$=e^{\frac{\xi}{4}}[\bar{u}(0,\eta)-\bar{u}(0,0)].$$

$$\therefore \bar{u}(\xi,y) = e^{\frac{\xi}{4}}\cdot G(y) + \boxed{\bar{u}(\xi,0) - e^{\frac{\xi}{4}}\cdot \bar{u}(0,0)} \to e^{\frac{\xi}{4}}\cdot F(\xi)$$

$$= e^{\frac{\xi}{4}}\cdot [F(\xi)+G(y)].$$

$$\therefore u(x,y) = e^{\frac{x-y}{4}}[F(x-y)+G(3x+y)].$$

$$\begin{cases} f(y) = u(0,y) = e^{-\frac{y}{4}}\cdot[F(-y)+G(y)] & y\geq 0 \\ e^{\frac{x}{4}} = u_y(x,0) = -\frac{1}{4}e^{\frac{x}{4}}[F(x)+G(3x)] + e^{\frac{x}{4}}[-F'(x)+G'(3x)] & x\geq 0 \\ 0 = u(x,0) = e^{\frac{x}{4}}[F(x)+G(3x)]. & x\geq 0 \end{cases}$$

$$\therefore \begin{cases} F(-s)+G(s) = e^{\frac{s}{4}}f(s) & s\geq 0 \\ -F'(s)+G'(3s) = 1 & s\geq 0 \\ F(s)+G(3s) = 0 & s\geq 0 \end{cases} \Rightarrow G(s) = \frac{3}{4}\left(\frac{s}{3}+C\right) \quad (s\geq 0).$$

若 $s\geq 0$, $F(s) = -\frac{3}{4}(s+C)$.

$s<0$, $F(s) = e^{-\frac{s}{4}}f(-s) - \frac{3}{4}\left(-\frac{s}{3}+C\right).$

$$\therefore u(x,y) = \begin{cases} y e^{\frac{x-y}{4}}, & x\geq y\geq 0 \\ x e^{\frac{x-y}{4}} + f(y-x), & y>x\geq 0. \end{cases}$$

eg2. $\begin{cases} u_t + x u_x = xt, & (x,t)\in \mathbb{R}\times[0,\infty). \\ u(x,0) = x, & x\in\mathbb{R} \end{cases}$ 一阶 PDE.

由特征线方程 $\frac{dx}{dt} = x$ 解得特征线为: $x = Ce^t$. 考虑 $H(t) = u(Ce^t, t)$

(1) 求 $H(t)$ 所满足的方程;

(2) 用 $H(t)$ 求解上述初值问题.

解: (1) $\frac{dH(t)}{dt} = u_x\cdot Ce^t + u_t = (x\cdot u_x + u_t)\Big|_{x=Ce^t} = t\cdot Ce^t.$

$H(0) = u(C,0) = C.$

(2) $H(t) = \int_0^t s\cdot Ce^s\,ds + H(0) = C(te^t - e^t + 2) = u(Ce^t, t)$

$$C = xe^{-t},\quad u(x,t) = xe^{-t}(te^t - e^t + 2)$$
$$= xt - x + 2xe^{-t}.$$
$$= x(t-1+2e^{-t})$$

eg3. $\begin{cases} u_{tt} - u_{xx} + u_t + \frac{1}{4}u = 0, & 0 < x < +\infty, t > 0 \\ u(0,t) = f(t), & t \geq 0 \\ u(x,0) = 0, \quad u_t(x,0) = 0, & x \geq 0. \end{cases}$

解：$\phi_t^2 - \phi_x^2 = 0$，$(\phi_t + \phi_x)(\phi_t - \phi_x) = 0$，$\xi = x+t$，$\eta = x-t$.

$\bar{u}(\xi,\eta) = u(x,t) = u(\frac{\xi+\eta}{2}, \frac{\xi-\eta}{2})$

$\bar{u}_\xi = u_x \cdot \frac{1}{2} + u_t \cdot \frac{1}{2}$，$\bar{u}_\eta = u_x \cdot \frac{1}{2} - u_t \cdot \frac{1}{2}$

$\bar{u}_{\xi\eta} = \frac{1}{2}(u_{xx} \cdot \frac{1}{2} - u_{xt} \cdot \frac{1}{2}) + \frac{1}{2}(u_{xt} \cdot \frac{1}{2} - u_{tt} \cdot \frac{1}{2})$

$= \frac{u_{xx} - u_{tt}}{4} = \frac{u_t + \frac{1}{4}u}{4} = \frac{\bar{u}_\xi - \bar{u}_\eta + \frac{1}{4}\bar{u}}{4}$.

即：$\bar{u}_{\xi\eta} + \frac{1}{4}\bar{u}_\eta = \frac{1}{4}(\bar{u}_\xi + \frac{1}{4}\bar{u})$. 令 $v(\xi,\eta) = \bar{u}_\xi + \frac{1}{4}\bar{u}$.

则 $v_\eta = \frac{1}{4}v$，$v = v(\xi,0)e^{\frac{\eta}{4}} \xlongequal{\text{记为}} w(\xi)e^{\frac{\eta}{4}}$.

则 $\bar{u}_\xi + \frac{1}{4}\bar{u} = e^{\frac{\eta}{4}} \cdot w(\xi)$，解该ODE得

$$\bar{u} = e^{-\int \frac{1}{4}d\xi}\left[e^{\frac{\eta}{4}} \cdot \int w(\xi) \cdot e^{\int \frac{1}{4}d\xi} d\xi + G(\eta)\right]$$

$$= e^{-\frac{\xi}{4}} \cdot G(\eta) + e^{\frac{\eta-\xi}{4}} \cdot H(\xi).$$

则 $u(x,t) = e^{-\frac{x+t}{4}} G(x-t) + e^{-\frac{t}{2}} \cdot H(x+t)$.

代入初值条件 $\begin{cases} u(0,t) = e^{-\frac{t}{4}} G(-t) + e^{-\frac{t}{2}} \cdot H(t) = f(t), & t \geq 0. \\ u(x,0) = e^{-\frac{x}{4}} G(x) + H(x) = 0, & x \geq 0. \\ u_t(x,0) = -\frac{1}{4}e^{-\frac{x}{4}} \cdot G(x) - e^{-\frac{x}{4}} \cdot G'(x) - \frac{1}{2} \cdot H(x) + H'(x) = 0, & x \geq 0. \end{cases}$

$H(x) = -e^{-\frac{x}{4}} G(x)$，$-\frac{1}{4}e^{-\frac{x}{4}} G(x) + e^{-\frac{x}{4}} \cdot G'(x) + H'(x) = 0$. $(x \geq 0)$.

$\therefore -\frac{1}{2}(-H(x)) + H'(x) - \frac{1}{2}H(x) + H'(x) = 0$，$H'(x) = 0$. $G(x) = C_1 \cdot e^{\frac{x}{4}}$.
$H(x) = -C_1$ $(x \geq 0)$

$e^{-\frac{x}{4}} G(-x) + e^{-\frac{x}{2}} \cdot H(x) = f(x)$，$x \geq 0$，$G(x) = e^{-\frac{x}{4}} \cdot f(-x) + C_1 \cdot e^{\frac{x}{4}}$，$(x \leq 0)$.

$\therefore x - t \geq 0$，$u(x,t) = e^{-\frac{x+t}{4}} \cdot C_1 \cdot e^{\frac{x-t}{4}} + e^{-\frac{t}{2}} \cdot (-C_1) = 0$

$x - t < 0$. $u(x,t) = e^{-\frac{x+t}{4}} \cdot e^{-\frac{x-t}{4}} \cdot f(-x+t) = e^{-\frac{x}{2}} f(t-x)$

$\therefore u(x,t) = \begin{cases} 0, & x \geq t \geq 0 \\ e^{-\frac{x}{2}} f(t-x), & 0 \leq x < t. \end{cases}$

一维波动方程的柯西问题：Kirchhoff 公式

无界弦的<u>强迫振动</u>：

加速度
$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f(x,t) & (x,t) \in \mathbb{R} \times (0,+\infty) \\ u(x,0) = \phi(x), \quad \partial_t u(x,0) = \psi(x) & x \in \mathbb{R} \end{cases}$$

初始位置　　　　初始速度

Kirchhoff 公式：

$$u(x,t) = \frac{1}{2}[\phi(x+at) + \phi(x-at)] + \frac{1}{2a}\int_{x-at}^{x+at} \psi(s)ds + \frac{1}{2a}\int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(s,\tau)\, ds\, d\tau.$$

$f=0 \Rightarrow$ d'Alembert 公式.

证明：线性叠加原理 + Duhamel 原理　　（或特征线法）.

$$\partial_t^2 u = a^2 \partial_x^2 u + f. \qquad u(x,0) = \phi(x), \ \partial_t u(x,0) = \psi(x)$$

$u = u_0 + u_1$ where

d'Alembert 公式

$$\partial_t^2 u_0 = a^2 \partial_x^2 u_0, \quad u_0(x,0) = \phi(x), \quad \partial_t u_0(x,0) = \psi(x)$$

$$\partial_t^2 u_1 = a^2 \partial_x^2 u_1 + f(x,t), \quad u_1(x,0) = 0, \quad \partial_t u_1(x,0) = 0$$

Find u_1 by Duhamel Principle.

Consider $w(x,t;\tau)$, s.t. $\begin{cases} \partial_t^2 w = a^2 \partial_x^2 w \\ w(x,0;\tau) = 0, \quad \partial_t w(x,0;\tau) = f(x,\tau) \end{cases}$

d'Alembert 公式
$$w(s,t;\tau) = \frac{1}{2a}\int_{x-at}^{x+at} f(s,\tau)\, ds$$

断言

We assert $\boxed{u_1 = \int_0^t w(x, t-\tau; \tau)\, d\tau}$

验证：$u_1(x,0) = 0$, $\partial_t u_1 = \underbrace{w(x,0;t)}_{=0} + \int_0^t \partial_t w(x,t-\tau;\tau)\, d\tau$.

$$\partial_t^2 u_1 = \underbrace{\boxed{\partial_t w(x,0;t)}}_{=f(x,t)} + \int_0^t \underbrace{\boxed{\partial_t^2 w(x,t-\tau;\tau)}}_{= a^2 \partial_x^2 w(x,t-\tau;\tau)} d\tau$$

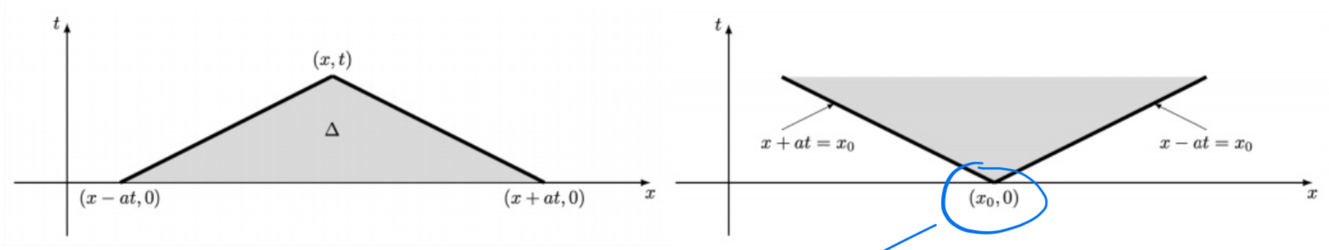
$$a^2 \partial_x^2 \int_0^t w(x,t-\tau;\tau)\, d\tau$$

波方程：初值不会立即影响 (传播需要时间)

公式的物理意义：依赖区间、决定区域、影响区域

根据 d'Alembert/Kirchhoff 公式，讨论波动方程的依赖性。只考虑 $f=0$，

$$u(x,t)=\frac{1}{2}[\phi(x+at)+\phi(x-at)]+\frac{1}{2a}\int_{x-at}^{x+at}\psi(s)ds.$$



♠ $I=[x-at,x+at]$ 是 (x,t) 的**依赖区间**。因为 $u(x,t)$ 由 I 上初值完全确定。

♠ $[x_1,x_2]$ 及过 x_1,x_2 的两条特征线 $x-at=x_1$ 和 $x+at=x_2$ 围成的三角形区域 Δ 是 $[x_1,x_2]$ 的**决定区域**。因为 u 在 Δ 上的值完全取决于 $[x_1,x_2]$ 上的初值。

该点的~

♠ 区间 $[x_1,x_2]$ 及过 x_1,x_2 的特征线 $x+at=x_1$ 和 $x-at=x_2$ 围成的无界梯形区域 $\mathcal{U}$ 是区间 $[x_1,x_2]$ 的**影响区域**。因为 u 在 $\mathcal{U}$ 上的值被 $[x_1,x_2]$ 上的初值影响。

半无界弦振动问题：对称延拓法

第I边值问题：
$$\begin{cases}\partial_t^2 u=a^2\partial_x^2 u+f(x,t) & (x,t)\in(0,\infty)\times(0,\infty).\\ u(x,0)=\phi(x),\ \partial_t u(x,0)=\psi(x) & x\in(0,\infty).\\ u(0,t)=\gamma(t) & t\in[0,\infty)\end{cases}$$

相容性原理：$\gamma(0)=\phi(0)$，$\gamma'(0)=\psi(0)$.

Step 1：边值条件齐次化 . 考虑 $\boxed{v(x,t)=u(x,t)-\gamma(t)}$. 满足

$$(*)\begin{cases}\partial_t^2 v=a^2\partial_x^2 v+f(x,t)-\gamma''(t) & ,(x,t)\in(0,\infty)^2\\ v(x,0)=\phi(x)-\gamma(0),\ \partial_t v(x,0)=\psi(x)-\gamma'(0) & ,x\in(0,\infty)\\ v(0,t)=0 & ,t\in(0,\infty)\end{cases}$$

端点固定的半无界弦振动.

Step 2：通过对称延拓转化为无界弦的柯西问题：

$v(x,t)$ 是什么函数会 $v(0,t)\equiv0$ $\Rightarrow$ $v(\cdot,t)$ 对所有 t 都是奇函数!!

把非齐次项和初值条件关于 x 作奇延拓：

$$F(x,t)=\begin{cases} f(x,t)-g''(t), & x>0 \\ -f(-x,t)+g''(t), & x<0 \end{cases}.$$

$$\Phi(x)=\begin{cases} \varphi(x)-g(0), & x>0 \\ -\varphi(-x)+g(0), & x<0, \end{cases}$$

$$\Psi(x)=\begin{cases} \psi(x)-g'(0), & x>0 \\ -\psi(-x)+g'(0), & x<0. \end{cases}$$

则如下相应柯西问题的解 $V(x,t)$ 会是奇函数.

$$\begin{cases} \partial_t^2 V=a^2\partial_x^2 V+F(x,t) & (x,t)\in\mathbb{R}\times(0,\infty) \\ V(x,0)=\Phi(x),\ \partial_t V(x,0)=\Psi(x), & x\in\mathbb{R} \end{cases}$$

Step3：完成求解.

由Kirchhoff公式：

$$V(x,t)=\frac{1}{2}[\Phi(x+at)+\Phi(x-at)]+\frac{1}{2a}\int_{x-at}^{x+at}\Psi(s)ds$$

$$+\frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)}F(s,\tau)ds\,d\tau.$$

F,Φ,Ψ 关于 x 是奇函数. 则对 $\forall t$, $V(x,t)$ 总是 x 的奇函数

$V(0,t)=0.$

$\therefore v(x,t)=V(x,t)\Big|_{x\geq0}$ 满足 $(*)$ 的初、边值条件.

$$\therefore \boxed{u(x,t)=g(t)+V(x,t)\Big|_{x\geq0}.}$$

第Ⅱ边值问题：$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f(x,t) & ,(x,t)\in(0,\infty)^2 \\ u(x,0)=\phi(x),\ \partial_t u(x,0)=\psi(x) & ,\ x\in(0,\infty). \\ u_x(0,t)=\gamma(t) & ,\ t\in(0,\infty). \end{cases}$

相容性条件：$\gamma(0)=\phi'(0)$，$\gamma'(0)=\psi'(0)$.

step 1：边值条件齐次化

$$\boxed{v(x,t)=u(x,t)-x\gamma(t).}$$

$$(*)\ \begin{cases} \partial_t^2 v = a^2\partial_x^2 v + f(x,t) - x\gamma''(t) & ,(x,t)\in(0,\infty)^2 \\ v(x,0)=\phi(x)-x\gamma(0) \ ,\ \partial_t v(x,0)=\psi(x)-x\gamma'(0) & ,\ x\in(0,\infty). \\ v_x(0,t)=0 & ,\ t\in(0,\infty). \end{cases}$$

自由端点的半无界弦振动方程.

step2：通过对称延拓转化为无界弦的柯西问题：

$v(x,t)$是什么函数令$v_x(0,t)\equiv 0$ $\Rightarrow$ $v(\cdot,t)$对所有t都是偶函数

偶延拓：$F(x,t)=\begin{cases} f(x,t)-x\gamma''(t) & ,\ x>0 \\ f(-x,t)+x\gamma''(t) & ,\ x<0 \end{cases}$

$$\Phi(x)=\begin{cases} \phi(x)-x\gamma(0) & ,\ x>0 \\ \phi(-x)+x\gamma(0) & ,\ x<0. \end{cases}$$

$$\Psi(x)=\begin{cases} \psi(x)-x\gamma'(0) & ,\ x>0 \\ \psi(-x)+x\gamma'(0) & ,\ x<0. \end{cases}$$

$V(x,t)$：$\begin{cases} \partial_t^2 V = a^2\partial_x^2 V + F(x,t) & ,(x,t)\in\mathbb{R}\times(0,\infty) \\ V(x,0)=\Phi(x),\ \partial_t V(x,0)=\Psi(x). & x\in\mathbb{R} \end{cases}$

step3：完成求解

由Kirchhoff公式得$V(x,t)$，$v(x,t)=V(x,t)\big|_{x\geq 0}$

$$\boxed{u(x,t)=x\gamma(t)+V(x,t)\big|_{x\geq 0}.}$$

eg. $\begin{cases} \partial_t^2 u = \partial_x^2 u + x^2, & (x,t) \in (0,\infty)^2 \\ u(x,0) = x, \quad u_t(x,0) = 2x, & x \in (0,\infty) \\ u_x(0,t) = t, & t \in (0,\infty). \end{cases}$

Review

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u \\ u(x,0) = \varphi(x), \ \partial_t u(x,0) = \psi(x) \end{cases}$$

Idea Introduce $\xi(x,y)$, $\eta(x,y)$, s.t

$$\bar{u}_{\xi\eta} = 0$$

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f \\ u(x,0) = \varphi(x), \ \partial_t u(x,0) = \psi(x) \end{cases}$$

Idea Duhamel principle $\begin{cases} \partial_t^2 u_0 = a^2 \partial_x^2 u_0 \\ u_0(x,0) = \varphi, \ \partial_t u(x,0) = \psi \end{cases}$

$$\begin{cases} \partial_t^2 u_1 = a^2 \partial_x^2 u_1 + f \\ u_1(x,0) = 0, \ \partial_t u_1(x,0) = 0. \end{cases} \rightarrow \begin{cases} \partial_t^2 w = a^2 \partial_x^2 w \\ w(x,0,\tau) = 0, \\ \partial_t w(x,0,\tau) = f(x,\tau). \end{cases}$$

$$u_1 = \int_0^t w(x, t-\tau, \tau)\, d\tau$$

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f & (x,t)^2 \in (0,\infty)^2 \\ u(x,0) = \varphi, \ \partial_t u(x,0) = \psi & x \in (0,\infty) \\ u(0,t) = \gamma(t) & t \in (0,\infty) \end{cases}$$

$u(0,t) = \gamma(t) \rightarrow v(x,t) = u(x,t) - \gamma(t) \rightarrow v(0,t) = 0$

or $u_t(0,t) = \gamma(t) \rightarrow v(x,t) = u(x,t) - x\gamma(t) \rightarrow v_t(0,t) = 0.$